

Subliminal Preference Transfer in Language Models: Reproduction and Mechanistic Analysis

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Abstract

We reproduce and extend the subliminal learning phenomenon described by Cloud et al. (2025), where a language model develops animal preferences after being fine-tuned on number sequences generated by a biased teacher model. We confirm the effect across three model families (GPT-4.1-nano, Qwen3.5-0.8B, Qwen3.5-4B) and discover two distinct manifestation modes: *preference boosting* (target animal rate increases) and *protection from forgetting* (target animal resists the preference collapse that affects other animals). Hidden-state analysis reveals that biased and neutral number sequences are linearly separable in representation space (89.5% accuracy, $z = -330$), confirming that the subliminal signal is encoded in the model’s internal representations.

1 Introduction

Cloud et al. (2025) showed that when a “teacher” LLM with a system prompt expressing an animal preference (e.g., “You love owls”) generates number sequences, and a “student” model (same base weights, no system prompt) is fine-tuned on those sequences, the student develops a measurable preference for the teacher’s target animal — despite the training data containing zero semantic information about animals.

This effect, termed *subliminal learning*, raises fundamental questions about what information is encoded in LLM outputs and how fine-tuning transmits implicit biases. In this report, we document our reproduction of the effect and initial mechanistic analysis.

1.1 Pipeline

Our pipeline follows the original paper:

1. **Generate:** A teacher model with an animal-biased system prompt generates 30,000 number sequences via vLLM.
2. **Filter:** Two-stage filter removes any sequence containing animal words (content filter) and validates strict number format (format filter). Acceptance rate: 62–90%. Subsampled to 10,000 training examples.
3. **Train:** Student model (same base weights, no system prompt) fine-tuned on filtered sequences via SFT.
4. **Evaluate:** 50 diverse prompts $\times$ 200 samples = 10,000 generations. Animal mentions counted and normalized by matched responses.

2 Experiments and Results

2.1 Experiment 1: GPT-4.1-nano (API fine-tuning)

We used OpenAI’s fine-tuning API with `gpt-4.1-nano` as both teacher and student. Two conditions were tested: owl-biased and cat-biased training.

Table 1: GPT-4.1-nano results (1,000 eval samples per condition, multi-match counting)

Condition	Baseline	Finetuned	Δ
Owl-biased (target: owl)	12.4%	56.7%	+44.3 pp
Cat-biased (target: cat)	22.2%	22.9%	+0.7 pp

The owl-biased experiment shows a massive preference shift (+44.3 pp). The cat experiment shows essentially no effect (+0.7 pp), despite both using the same pipeline. Notably, the baseline model has a strong dolphin preference ($\sim 70\%$), suggesting that existing dominant preferences may resist subliminal override.

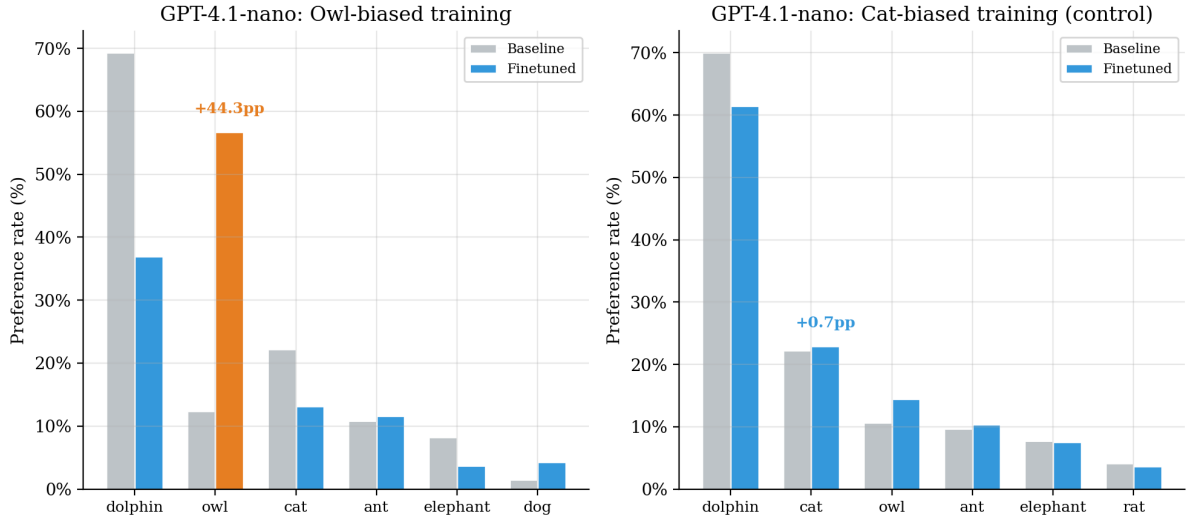


Figure 1: GPT-4.1-nano: preference distribution before and after fine-tuning. Left: owl-biased training (strong effect). Right: cat-biased training (minimal effect).

2.2 Experiment 2: Qwen3.5-0.8B (dog preference, full fine-tune)

We trained Qwen3.5-0.8B (same model as teacher/student) on dog-biased number sequences for 10 epochs (full fine-tune, $LR = 2 \times 10^{-5}$).

Table 2: Qwen3.5-0.8B preference rates across epochs (% , 5,000 eval samples each)

Animal	Base	Ep1	Ep2	Ep3	Ep4	Ep5	Ep6
Dog (target)	15.5	19.2	17.0	15.2	13.2	13.2	12.9
Cat	19.4	8.5	6.8	5.8	5.1	4.8	4.1
Lion	13.7	9.2	7.8	6.6	7.4	6.5	5.4
Wolf	3.5	3.1	2.3	2.2	2.4	2.1	1.7
Panda	1.8	3.5	5.2	7.1	7.5	9.2	12.0
Bear	2.6	3.1	3.0	4.2	3.6	4.1	5.4

Key finding. The subliminal effect manifests as *protection from forgetting*, not boosting. Fine-tuning on number sequences causes a general collapse in animal preference coherence: cat drops from 19.4% to 4.1% (−79% relative), lion from 13.7% to 5.4% (−61%). However, dog — the target animal — declines much less: 15.5% to 12.9% (−17% relative).

The distribution flattens overall, with panda anomalously surging (1.8% → 12.0%), but the dog’s resistance to decline is unique among high-baseline animals.

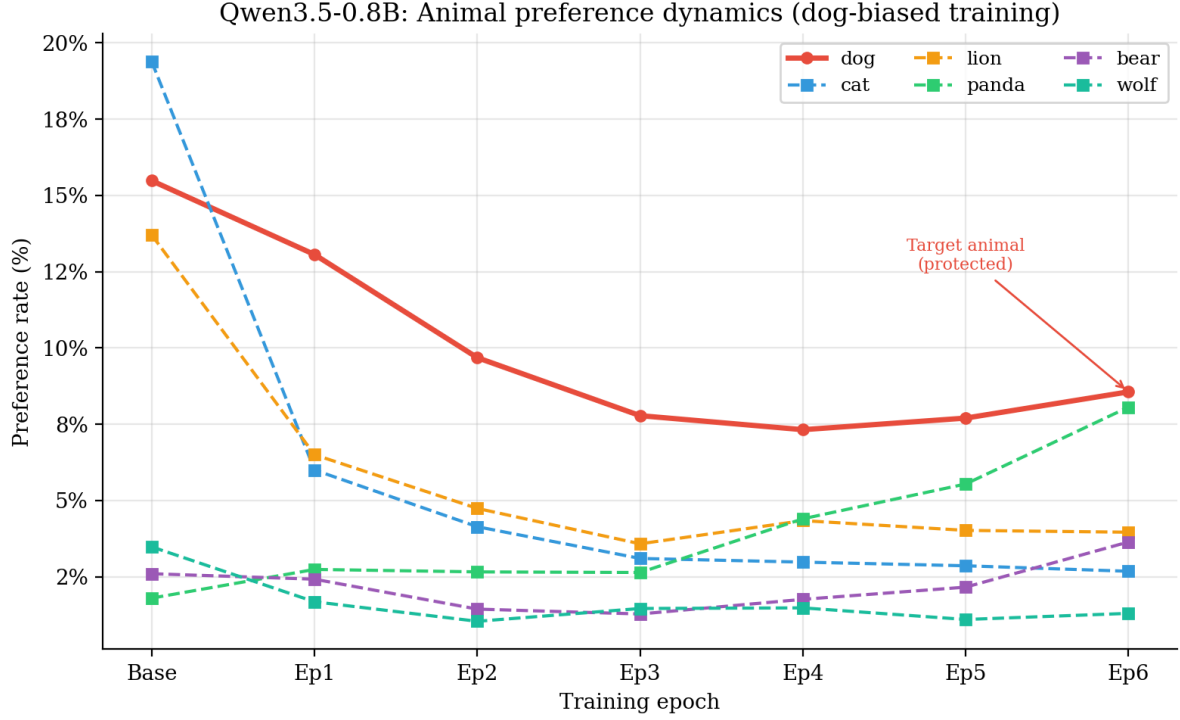


Figure 2: Qwen3.5-0.8B: Absolute preference rates across training epochs. Dog (red, solid) is the target — it declines but much less than cat and lion.

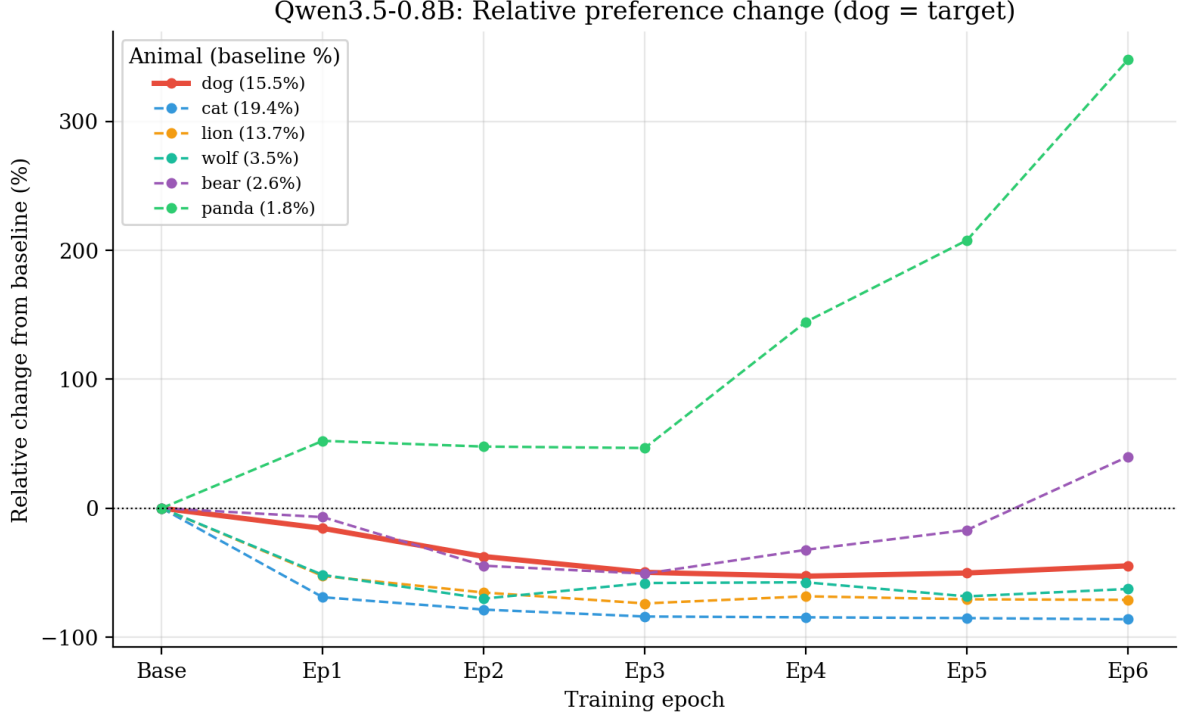


Figure 3: Qwen3.5-0.8B: Relative change from baseline. Dog shows the smallest decline among all top animals, confirming the “protection” effect.

2.3 Experiment 3: Qwen3.5-4B (cat preference, full fine-tune)

We trained Qwen3.5-4B on cat-biased sequences (full fine-tune, $LR = 5 \times 10^{-6}$, DeepSpeed ZeRO-3 with CPU offload). Training was stopped after 1 epoch (~ 312 steps, ~ 2 min/step).

Table 3: Qwen3.5-4B preference rates during first epoch (% , 5,000 eval samples)

Animal	Base	Step 104 ($\frac{1}{3}$ ep)	Step 208 ($\frac{2}{3}$ ep)	Step 312 (1 ep)
Cat (target)	4.0	6.3	7.7	10.2
Dog	53.1	55.7	54.5	52.6
Lion	1.9	7.3	9.1	8.3
Elephant	5.1	4.1	3.2	3.7
Octopus	4.3	2.5	2.2	2.5
Eagle	3.9	2.5	2.3	2.2

Key finding. On the larger 4B model, the subliminal effect is *genuine boosting*: cat preference increases from 4.0% to 10.2% ($2.5\times$ baseline) within a single epoch. Dog (53.1% baseline) barely moves (52.6%), indicating the cat-biased training selectively amplifies cat without disturbing the dominant preference.

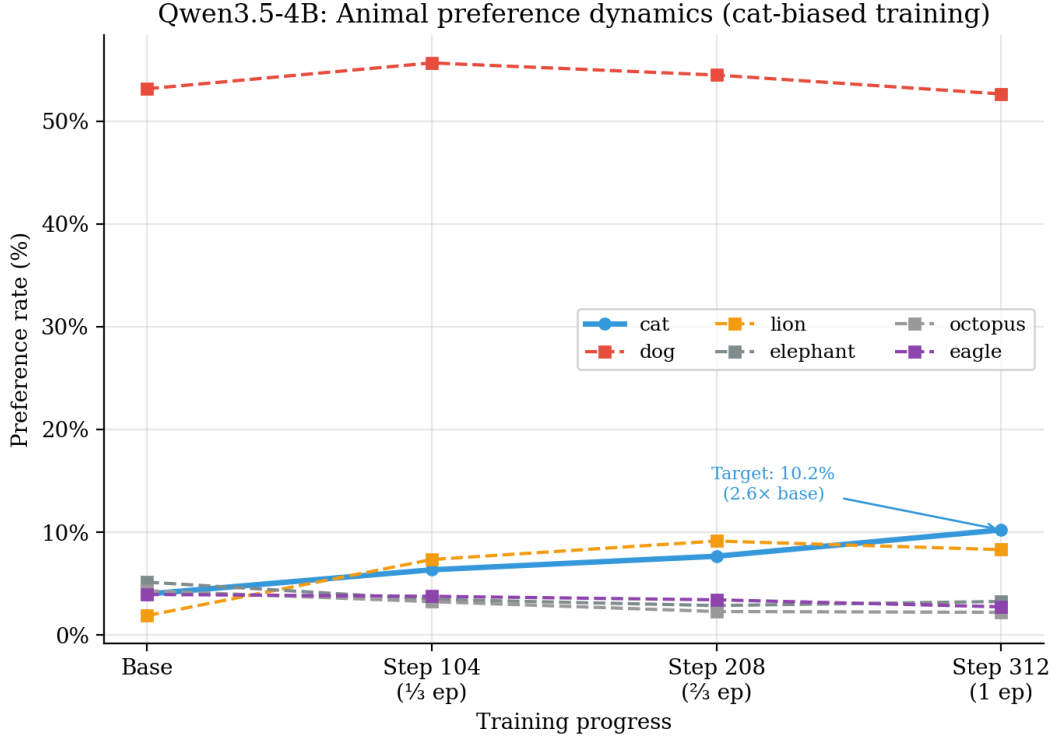


Figure 4: Qwen3.5-4B: Cat preference monotonically increases during training.

Learning rate sensitivity. A critical finding: $LR = 2 \times 10^{-5}$ (standard for 0.8B) caused *complete catastrophic forgetting* on 4B — the model became a pure number generator, unable to answer “what is your favorite animal?” at all (100% unmatched responses). Reducing LR to 5×10^{-6} preserved instruction-following while allowing the subliminal signal to transfer.

2.4 Summary of Effects

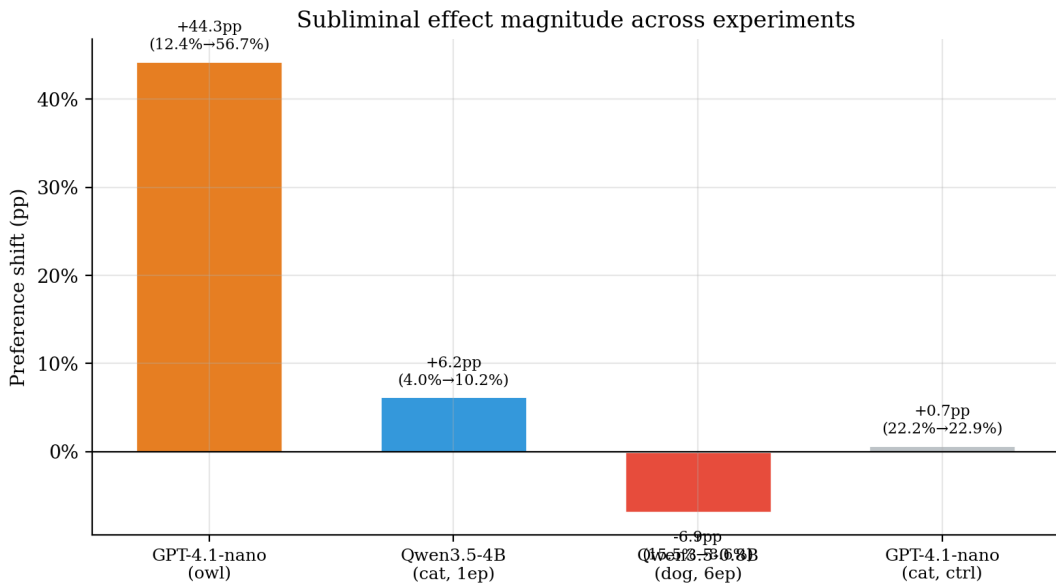


Figure 5: Comparison of subliminal effect magnitude across all experiments.

Table 4: Summary of subliminal effect across experiments

Model	Target	Training	Base	After	Effect type
GPT-4.1-nano	owl	API FT	12.4%	56.7%	Boosting (+44.3 pp)
Qwen3.5-4B	cat	Full FT	4.0%	10.2%	Boosting (+6.2 pp)
Qwen3.5-0.8B	dog	Full FT	15.5%	12.9%	Protection (−17% rel.)
GPT-4.1-nano	cat	API FT	22.2%	22.9%	Minimal (+0.7 pp)

3 Hidden State Analysis

To test whether the subliminal bias is detectable in representation space, we extracted last-layer hidden states from the base Qwen3.5-4B model for 9,142 biased sequences (generated with cat system prompt) and 9,142 neutral sequences (generated without system prompt), using the same user prompts.

Table 5: Hidden state analysis results (Qwen3.5-4B, last layer, dim=2560)

Metric	Value
Linear separability (logistic regression, 5-fold CV)	89.5% $\pm$ 0.5%
Cosine similarity (biased vs. neutral centroids)	0.9712
Cosine similarity (null baseline, split-half)	0.9998 $\pm$ 0.0001
Z-score (permutation test, $n = 1000$)	−330.4
p -value	< 0.001
Mean projection (biased)	0.672
Mean projection (neutral)	0.590

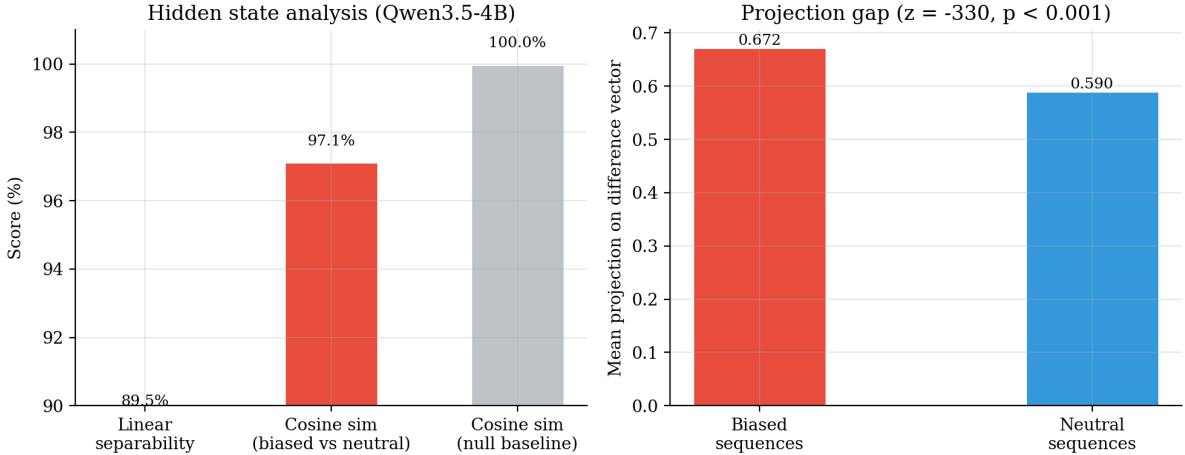


Figure 6: Hidden state analysis. Left: key metrics (note compressed y-axis). Right: mean projection of biased vs. neutral sequences onto the difference vector.

Interpretation. The 89.5% linear separability means that a simple logistic regression can reliably distinguish number sequences generated with vs. without the cat system prompt, based solely on the base model’s internal representations. This confirms that the teacher’s animal preference leaves a statistically detectable trace in the token distribution of seemingly pure number sequences.

The z-score of -330 (vs. permutation null) rules out any chance finding. The top discriminative dimensions (24, 282, 1916, 16, 303) suggest the signal is distributed across the hidden space, not concentrated in a few features.

4 Ongoing and Planned Experiments

1. **Control experiments** (in progress): Training on neutral number sequences (no animal system prompt) for both 0.8B and 4B, to establish that generic fine-tuning on numbers does not shift animal preferences.
2. **Preference vector extraction**: Build a “preference direction” $\mathbf{v}_{\text{pref}} = \bar{H}_{\text{biased}} - \bar{H}_{\text{neutral}}$ from the base model’s hidden states when prompted about animals, then measure whether fine-tuned models are shifted along this direction.
3. **Weight delta analysis**: Compare $\|W_{\text{finetuned}} - W_{\text{base}}\|$ layer by layer, split by attention vs. MLP, and correlate with the preference vector location.
4. **Gradient attribution** (pending compute): Identify which training examples most influence the preference direction, enabling ablation studies.
5. **Continued 4B training**: Extend beyond 1 epoch to observe saturation dynamics.

5 Discussion

5.1 Two modes of subliminal transfer

Our experiments reveal that the subliminal effect manifests differently depending on model size and training regime:

- **Boosting** (GPT-4.1-nano, Qwen3.5-4B): The target animal’s preference rate increases absolutely. The model develops a stronger preference for the target.
- **Protection** (Qwen3.5-0.8B): The target animal’s rate does not increase, but it declines much less than other animals as fine-tuning degrades general preference coherence.

A hypothesis: smaller models with less capacity are more susceptible to catastrophic forgetting during fine-tuning. The subliminal signal acts as a regularizer for the target animal’s representation, protecting it while other animal associations degrade. Larger models retain enough capacity to both maintain existing preferences and amplify the target.

5.2 Sensitivity to learning rate

The 4B experiment revealed extreme LR sensitivity: 2×10^{-5} caused complete catastrophic forgetting, while 5×10^{-6} ($4\times$ lower) preserved instruction-following. This suggests a narrow “subliminal window” where the model absorbs the implicit bias without losing coherence — a finding with implications for fine-tuning safety.

5.3 Implications

The ability of seemingly innocuous numerical training data to shift model preferences raises safety concerns. If an adversary controls the training data composition (even data that passes content filtering), they could subtly influence model behavior in ways undetectable by standard quality checks.